Chat Link: https://parley.mit.edu/share/1njavmuyi_qCTjrp56Dyu

```
global_data = {

    # ─── ATTRIBUTE LABELS ──────────────────────────────
    "attribute_1": "total_cost",
    "attribute_2": "material_continuity",
    "attribute_3": "mass_kg_per_m",
    "attribute_4": "manufacturing_joining_compatibility",
    "attribute_5": "dc_resistance_ohm_per_km",

    # ─── DIRECTIONS ────────────────────────────────────
    "direction_attribute_1": "negative",  # lower cost is better
    "direction_attribute_2": "positive",  # higher material continuity is better
    "direction_attribute_3": "negative",  # lower mass is better
    "direction_attribute_4": "positive",  # higher compatibility is better
    "direction_attribute_5": "negative",  # lower resistance is better

    # ─── NORMALIZATION BOUNDS ──────────────────────────
    # Bounds are set to cover the FULL PHYSICAL RANGE of this product family
    # (residential trunk line / service entrance cable, 6 AWG – 400 kcmil,
    #  Cu or Al, any global market), not just the current 8-product dataset.

    "attribute_1_min": 1.00,    # USD/m  — floor below smallest Al SER in cheapest market
    "attribute_1_max": 80.00,   # USD/m  — ceiling above largest Cu SER in highest-labor market

    "attribute_2_min": 0.00,    # absolute logical floor (complete material discontinuity)
    "attribute_2_max": 1.00,    # absolute logical ceiling (perfect historical match)

    "attribute_3_min": 0.10,    # kg/m   — below smallest 6 AWG Al SER (~0.26 kg/m)
    "attribute_3_max": 8.00,    # kg/m   — above largest 400 kcmil Cu SER (~6.25 kg/m)

    "attribute_4_min": 0.00,    # absolute logical floor (complete incompatibility)
    "attribute_4_max": 1.00,    # absolute logical ceiling (perfect compatibility)

    "attribute_5_min": 0.05,    # Ω/km   — below 400 kcmil Cu (~0.085 Ω/km)
    "attribute_5_max": 3.00,    # Ω/km   — above 6 AWG Al (~2.12 Ω/km)

    # ─── COMMODITY PRICES ──────────────────────────────
    "copper_price_per_ton":   9500,
    "aluminum_price_per_ton": 2500,
    "copper_price_per_kg":   9.5,
    "aluminum_price_per_kg": 2.5,
}
```

```
regional_data = [

  # ── UNITED STATES ───────────
  # Aluminum has gained significant share in large-gauge residential feeders
  # driven by cost, but copper retains ~40% due to NEC termination culture,
  # historical preference, and single-family custom construction.
  {
    "region": "United States",
    "copper_product_market_share":   0.40,  # declining but stable in custom/single-family
    "aluminum_product_market_share": 0.60,  # majority in multi-family and cost-driven builds

    # Weights established in prior session (5-attribute framework):
    "weight_attribute_1": 3,  # total_cost          — moderate; NEC compliance prioritized
    "weight_attribute_2": 4,  # material_continuity — high; copper deeply entrenched
    "weight_attribute_3": 1,  # mass                — low; short runs, experienced crews
    "weight_attribute_4": 5,  # mfg/joining compat  — highest; NEC lug ratings are strict
    "weight_attribute_5": 2,  # dc_resistance       — low; code-compliant sizing equalizes
  },

  # ── CHINA ────────────────────────────
  # Aluminum dominates large-gauge residential building wire in China.
  # GB/T standards support both, but cost pressure drives >75% Al share.
  {
    "region": "China",
    "copper_product_market_share":   0.25,
    "aluminum_product_market_share": 0.75,

    "weight_attribute_1": 5,  # total_cost          — highest; cost is primary driver
    "weight_attribute_2": 3,  # material_continuity — moderate; Al incumbency growing
    "weight_attribute_3": 2,  # mass                — low-moderate; high-rise riser logistics
    "weight_attribute_4": 3,  # mfg/joining compat  — moderate; both materials supported
    "weight_attribute_5": 3,  # dc_resistance       — moderate; electrification load growth
  },

  # ── INDIA ───────────────────────────────────
  # Aluminum widely used for large residential feeders. IS 694/IS 1554 code
  # enforcement is variable; cost is the near-exclusive driver.
  {
    "region": "India",
    "copper_product_market_share":   0.20,
    "aluminum_product_market_share": 0.80,
```

```
    "weight_attribute_1": 5,  # total_cost          — highest; dominant procurement criterion
    "weight_attribute_2": 3,  # material_continuity — moderate; Al widely incumbent
    "weight_attribute_3": 2,  # mass                — low-moderate; dense urban construction
    "weight_attribute_4": 2,  # mfg/joining compat  — low; field enforcement inconsistent
    "weight_attribute_5": 3,  # dc_resistance       — moderate; growing electrification loads
  },

  # ── EUROPE ─────────────────────────────────────
  # Copper strongly preferred for residential wiring in EU. IEC 60228/EN
  # harmonized standards allow Al but Energy Performance of Buildings
  # Directive elevates electrical efficiency as a measurable design criterion.
  {
    "region": "Europe",
    "copper_product_market_share":   0.65,
    "aluminum_product_market_share": 0.35,

    "weight_attribute_1": 3,  # total_cost          — moderate; lifecycle TCO weighted
    "weight_attribute_2": 4,  # material_continuity — high; copper historical norm in EU
    "weight_attribute_3": 1,  # mass                — low; short residential runs
    "weight_attribute_4": 4,  # mfg/joining compat  — high; CE-marked hardware ecosystem
    "weight_attribute_5": 4,  # dc_resistance       — high; EPBD energy efficiency mandates
  },
]


product_data = [

  # ══════════════════════════════════════════
  # UNITED STATES
  # ══════════════════════════════════════════

  {
    "region": "United States",
    "dominant material": "cu",
    "product": "Copper Type SER 2/0 AWG Service Entrance Cable "
               "(3c XHHW-2 + 4 AWG Cu Bare Ground, NEC 200A, per meter)",

    # ── Conductor material mass (kg/m) ───────
    # 3 × 2/0 AWG Cu (67.4 mm² ea.) + 4 AWG Cu bare ground (21.2 mm²)
    # ρ_Cu = 8,960 kg/m³ → 3(67.4 × 10⁻⁶ × 8960) + (21.2 × 10⁻⁶ × 8960)
    "copper_kg":     2.003,
    "aluminum_kg":   0.000,

    # total = copper + PVC insulation/jacket (~0.277 kg/m)
```

```
    "total_mass_kg": 2.28,

    # non_material = insulation, extrusion, labor, overhead (US high-labor market)
    # Total cost = 2.003 × $9.50 + 0 × $2.50 + $8.00 = $27.03/m
    "non_material_cost_per_unit": 8.00,

    # Attribute values ──────────────────────────────────
    "attribute_2_value": 0.85,   # material_continuity: copper strongly in
    "attribute_3_value": 2.28,   # mass_kg_per_m: total cable weight
    "attribute_4_value": 0.95,   # mfg/joining compat: copper IS the NEC baseline
    "attribute_5_value": 0.255,  # dc_resistance Ω/km: 1.72e-8 / (67.4e-6) × 1000
},

{
    "region": "United States",
    "dominant material": "al",
    "product": "Aluminum Type SER 4/0 AWG Service Entrance Cable "
            "(3c Al AA-8000 + 4 AWG Cu Bare Ground, NEC 200A equiv., per meter)",

    # 3 × 4/0 AWG Al (107.2 mm² ea.) | ρ_Al = 2,700 kg/m³
    # → 3(107.2 × 10⁻⁶ × 2700) = 0.868 kg/m Al
    # + 4 AWG Cu ground: 21.2 × 10⁻⁶ × 8960 = 0.190 kg/m Cu
    "copper_kg":     0.190,
    "aluminum_kg":   0.868,

    # total = Al + Cu ground + PVC insulation/jacket (~0.332 kg/m)
    "total_mass_kg": 1.39,

    # Total cost = 0.190 × $9.50 + 0.868 × $2.50 + $8.00 = $11.98/m
    "non_material_cost_per_unit": 8.00,

    "attribute_2_value": 0.15,   # material_continuity: breaks from copper-dominant US history
    "attribute_3_value": 1.39,   # mass_kg_per_m: ~39% lighter than Cu equivalent
    "attribute_4_value": 0.40,   # mfg/joining compat: requires AL-rated lugs, anti-oxidant,

    "attribute_5_value": 0.263,  # dc_resistance Ω/km: 2.82e-8 / (107.2e-6) × 1000
},

# ══════════════════════════════════════════
# CHINA
# ══════════════════════════════════════════════

{
    "region": "China",
```

```
    "dominant material": "cu",
    "product": "Copper GB/T 70 mm² Service Entrance Feeder "
               "(3c XLPE + 4 AWG Cu Bare Ground, 200A, per meter)",

    # IEC/GB equivalent: 70 mm² ≈ 2/0 AWG (67.4 mm²) — same conductor mass basis
    "copper_kg":     2.003,
    "aluminum_kg":   0.000,
    "total_mass_kg": 2.28,

    # Total cost = $19.03 + $4.00 = $23.03/m (lower labor/overhead vs US)
    "non_material_cost_per_unit": 4.00,

    "attribute_2_value": 0.30,   # material_continuity: aluminum is growing incumbent in China
                                 # for large-gauge residential; copper not strongly dominant
    "attribute_3_value": 2.28,
    "attribute_4_value": 0.80,   # mfg/joining compat: GB standards support Cu well,
                                 # both termination ecosystems co-exist
    "attribute_5_value": 0.255,
},

{
    "region": "China",
    "dominant material": "al",
    "product": "Aluminum GB/T 120 mm² Service Entrance Feeder "
               "(3c Al + 4 AWG Cu Bare Ground, 200A equiv., per meter)",

    # IEC/GB equivalent: 120 mm² ≈ 4/0 AWG (107.2 mm²) — closest standard IEC size
    # Using same conductor mass basis as AWG equivalent for consistency
    "copper_kg":     0.190,
    "aluminum_kg":   0.868,
    "total_mass_kg": 1.39,

    # Total cost = $3.98 + $4.00 = $7.98/m
    "non_material_cost_per_unit": 4.00,

    "attribute_2_value": 0.70,   # material_continuity: aluminum is the emerging norm
                                 # for trunk lines in Chinese residential construction
    "attribute_3_value": 1.39,
    "attribute_4_value": 0.75,   # mfg/joining compat: Al widely accepted in GB/T;
                                 # AA-8000 alloy and compatible lugs broadly available
    "attribute_5_value": 0.263,
},

# ══════════════════════════════════════════════════════
```

```
# INDIA
# ════════════════════════════════════════════════════════════

{
    "region": "India",
    "dominant material": "cu",
    "product": "Copper IS 694 70 mm² Service Entrance Cable "
            "(3c PVC/XLPE + 4 AWG Cu Bare Ground, 200A, per meter)",

    "copper_kg":     2.003,
    "aluminum_kg":   0.000,
    "total_mass_kg": 2.28,

    # Total cost = $19.03 + $3.50 = $22.53/m
    "non_material_cost_per_unit": 3.50,

    "attribute_2_value": 0.25,   # material_continuity: aluminum heavily used in India
                                 # for large residential feeders; Cu not strongly incumbent
    "attribute_3_value": 2.28,
    "attribute_4_value": 0.70,   # mfg/joining compat: IS 694 supports Cu; infrastructure
                                 # adequate but termination practice variability reduces score
    "attribute_5_value": 0.255,
},

{
    "region": "India",
    "dominant material": "al",
    "product": "Aluminum IS 694 120 mm² Service Entrance Cable "
            "(3c Al + 4 AWG Cu Bare Ground, 200A equiv., per meter)",

    "copper_kg":     0.190,
    "aluminum_kg":   0.868,
    "total_mass_kg": 1.39,

    # Total cost = $3.98 + $3.50 = $7.48/m
    "non_material_cost_per_unit": 3.50,

    "attribute_2_value": 0.75,   # material_continuity: aluminum is the dominant incumbent
                                 # for residential trunk lines in India
    "attribute_3_value": 1.39,
    "attribute_4_value": 0.55,   # mfg/joining compat: Al accepted under IS 694 but
                                 # variable field practices and anti-oxidant awareness
                                 # reduce effective compatibility vs mature markets
    "attribute_5_value": 0.263,
```

```
},

# ════════════════════════════════════════
# EUROPE
# ════════════════════════════════════════════════════

{
    "region": "Europe",
    "dominant material": "cu",
    "product": "Copper IEC 60228 70 mm² Service Entrance Cable "
               "(3c XLPE/PVC + 4 AWG Cu Bare Ground, 200A, per meter)",

    "copper_kg":     2.003,
    "aluminum_kg":   0.000,
    "total_mass_kg": 2.28,

    # Total cost = $19.03 + $7.50 = $26.53/m
    "non_material_cost_per_unit": 7.50,

    "attribute_2_value": 0.80,   # material_continuity: copper is the strong residential
                                 # default across EU (slightly below US due to Al use in
                                 # some utility/MV feeder contexts)
    "attribute_3_value": 2.28,
    "attribute_4_value": 0.90,   # mfg/joining compat: IEC/CE-marked panel ecosystem
                                 # is built around copper for residential applications
    "attribute_5_value": 0.255,
},

{
    "region": "Europe",
    "dominant material": "al",
    "product": "Aluminum IEC 60228 120 mm² Service Entrance Cable "
               "(3c Al + 4 AWG Cu Bare Ground, 200A equiv., per meter)",

    "copper_kg":     0.190,
    "aluminum_kg":   0.868,
    "total_mass_kg": 1.39,

    # Total cost = $3.98 + $7.50 = $11.48/m
    "non_material_cost_per_unit": 7.50,

    "attribute_2_value": 0.20,   # material_continuity: breaks sharply from EU copper norm
    "attribute_3_value": 1.39,
    "attribute_4_value": 0.55,   # mfg/joining compat: IEC 61386 and harmonized EN
```

                                        # standards accept Al with correct connectors, but the
                                        # CE-marked residential hardware ecosystem defaults to Cu
        "attribute_5_value": 0.263,
    },
]